Mesiri

Field Guide · 03a · Deep dive inside Materials · Internal team edition

Material categories

Why a builder sorts hundreds of items into Structural, Electrical, Plumbing and the rest; what that sorting is used for; exactly how Mesiri does it in the database, the dashboard, WhatsApp and reports; and what to build next.

Draft, 8 Nov 2026. Parts 1 to 3 are about the business. Parts 4 to 8 describe Mesiri's code as it is on `main` today (PRC-01, 02, 02b, 02c, 03, 05, 17a to 17d). Business practices here should be checked by someone with site experience.

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1 · WHAT A MATERIAL CATEGORY IS

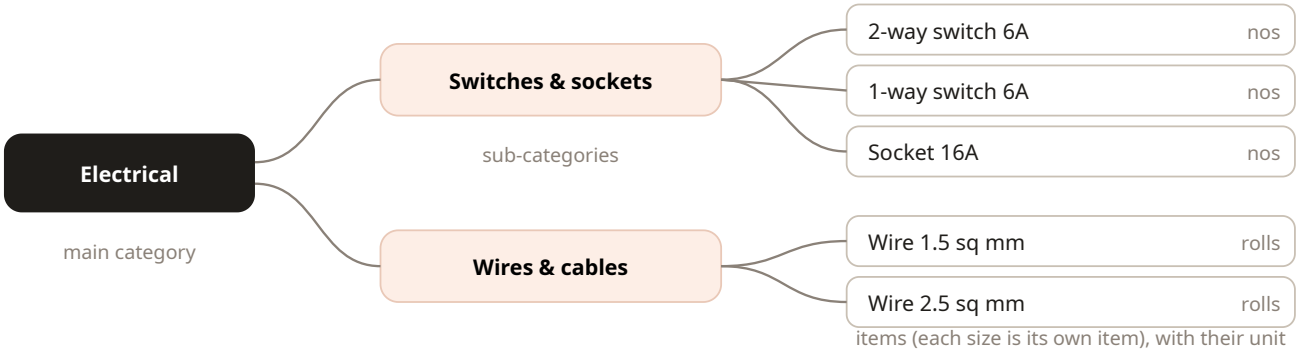
A shelf in the store, for hundreds of items

A mid-size building project uses a few hundred different items: cement and steel, but also 2-way switches, 20 mm elbows, 6A sockets, CP taps, tile spacers. A **category** is the shelf an item sits on, so people can find it, count it, total it and give it to the right person. It does not change what the item is; it groups it.

Four words that are easy to mix up

WORD	WHAT IT IS	EXAMPLE
Main category	The trade or big group.	Electrical
Sub-category	A group inside it.	Electrical › Switches & sockets
Item	One thing you buy, count and use. Each size or type is its own item.	2-way switch 6A · 1-way switch 6A
Unit	What one item is counted in.	nos, bags, cft, kg, rolls

The rule that keeps it simple: sizes and types are **items**, not categories. "2-way switch 6A" and "1-way switch 6A" are two items under one sub-category. If sizes became categories, the tree would grow to thousands of shelves with one item each.



2 · WHY IT IS USEFUL: TEN JOBS IT DOES

Sorting is not tidiness; it is how a builder controls what matters

#	JOB	WITHOUT CATEGORIES	WITH CATEGORIES
1	**See stock at a glance**	One list of 300 items; cement is lost among fittings.	"Structural: 5 items, 1 running low." Open only what you care about.
2	**Know where the money went**	"₹28 lakh on material this month." On what?	Structural ₹19 L, Electrical ₹4 L, Plumbing ₹3 L, Finishing ₹2 L.
3	**Ask by trade**	"Which electrical items are low?" means reading every row.	One question, one answer: "Electrical stock?"

#	JOB	WITHOUT CATEGORIES	WITH CATEGORIES
4	**Give work to the right person**	Everything goes to the GM.	The MEP engineer owns Electrical and Plumbing; the site engineer owns Structural.
5	**Buy from the right supplier**	Any shop for anything.	Electrical shop for Electrical, cement dealer for Structural; rates compared within a category.
6	**Readable reports**	The daily report lists 40 materials in random order.	The daily report groups them: Structural, then Electrical, then Finishing.
7	**Control by value**	Every item checked the same way.	Tight control of Structural (expensive), lighter on Hardware (cheap).
8	**Compare with the budget**	Material spend can't be matched to the estimate.	Electrical material vs the electrical part of the BOQ.
9	**Tax**	GST rate looked up item by item.	Similar items share a GST rate and HSN code (planned).
10	**Fast setup**	Every new company starts from an empty list.	A standard builder list on day one; import your own items into it.

> *"Every item sits under a main category (Structural, Electrical, Plumbing, Fire, Gas, Lift…), with sub-categories (Switches, Pipes, CP fittings) and sizes or types."*
>
> A builder customer, sales meeting, 5 Nov 2026 (why Mesiri built categories)

3 · HOW CONSTRUCTION COMPANIES SORT MATERIAL

By trade is the most common; companies change the names

SCHEME	LOOKS LIKE	WHEN IT IS USED
By trade (Mesiri's default)	Structural · Electrical · Plumbing & CP · Fire · Gas · Lift · Finishing	Most builders and contractors; matches who installs it and who sells it.
By work section	Earthwork · Concrete · Masonry · Plastering · Flooring · Painting	Companies that think in BOQ sections; good for cost comparison.
By stage	Foundation · Superstructure · Finishing · External works	Developers planning cash by stage.
By accounting group	Raw material · Consumables · Tools · Spares	Accounts and the CA; often in Tally.

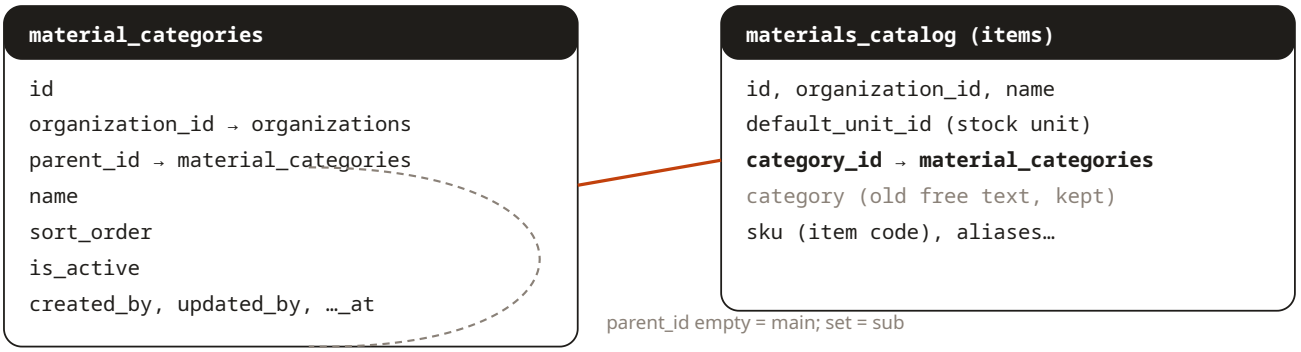
Trade is the best **starting point** because it matches three things at once: who supplies it, who installs it, and how people talk about it on site. A company can then rename, add and switch off categories to fit its own habits. The other schemes are usually needed **alongside** it (work section for cost, accounting group for the CA), which is why part 8 links categories to cost centres instead of replacing them.

Good rules for a category list

- **Two levels are enough** for a mid-size builder: main and sub. A third level is where lists become unusable.
- **One item, one place.** An item sits in exactly one category; reports never double count.
- **The company owns the list.** Rename "Plumbing & CP" to "Plumbing" if that is what they say.
- **Never delete a used category.** Switch it off: old items keep it, new items can't pick it.
- **"No category" is visible**, never hidden, so someone places those items.
- **Never guess.** A wrong category is worse than none: "Stainless steel sink" is not Steel, "Sand paper" is not Masonry.

4 · MESIRI'S MODEL: TABLES AND RULES

One table for the tree, one column on each item



RULE	HOW IT IS ENFORCED
Per company	Every row has `organization_id`; row-level security (as every company table) keeps one company from seeing another's.
Two levels	A row with no parent is a main category; a sub's parent must be a main one. The API refuses a sub under a sub ("already a sub-category; add this under its main category").
Unique names per parent	A unique index on (company, parent, lower-case name): "Pipes" can exist under Plumbing and under Fire protection, but not twice under Plumbing.
Switch off, never lose	`is_active = false`: items keep it, it is no longer offered. Delete works only for an empty category; otherwise 409 "move them out first, or switch it off".
Additive	Migration 0503 added the table and a nullable `category_id`; the old free-text `category` column was kept, not rewritten, and is now only a hint.
Main includes subs	Filtering by a main category (stock, deliveries, usage) includes every sub under it.

The standard builder list (every new company gets it)

MAIN CATEGORY	SUB-CATEGORIES
Structural	Masonry · Concrete · Steel · Shuttering
Electrical	Wires & cables · Switches & sockets · Conduits & boxes · DB & MCBs · Lights & fittings
Plumbing & CP	Pipes · Pipe fittings · Valves · Sanitaryware · CP fittings & accessories · Tanks
Fire protection	Pipes & fittings · Sprinklers · Hydrants & hoses · Extinguishers · Alarms & detectors
Gas	Pipes & fittings · Regulators & valves
Lift	(none)
Finishing	Tiles & flooring · Paint & putty · Waterproofing · Doors & windows · Hardware

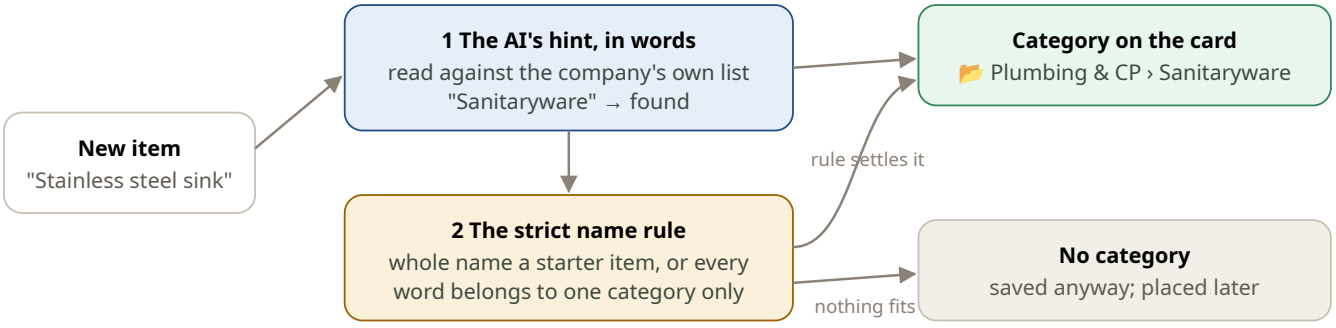
7 main, 27 sub, in the order a builder works through a project. Existing companies are never changed silently: they get the list with one tap ("Add the standard builder list"), safe to repeat.

5 · HOW A MATERIAL GETS ITS CATEGORY

Four ways in, and one rule above them: never guess

WAY	WHAT HAPPENS	EXAMPLE
1 Picked on the dashboard	Items tab: a category picker on each item; a "No category" list with a picker and the old free-text group shown as a hint.	Wash basin → Plumbing & CP › Sanitaryware
2 Excel import	Columns Main category, Sub-category, Item, Unit, Item code, Other names. New sub-categories are added; existing items without a category get the one in the file.	A 400-row item list loaded at setup
3 Standard list for existing items	When a company adds the standard list, items are placed only by a starter name or an exact old group; anything else stays under No category.	"Masonry" (old group) → Structural › Masonry
4 Suggested for a new item	When someone adds a new material (dashboard or WhatsApp), Mesiri suggests its category; it shows on the card before Save.	"MCB 32A" → Electrical › DB & MCBs

How the suggestion works (and why it is strict)



NAME	RESULT	WHY
Cement	Structural › Masonry	A starter item's name (the starter list files cement under Masonry)
Binding wire 18 gauge	Structural › Steel	A starter item; sizes ("18 gauge") are ignored
MCB 32A	Electrical › DB & MCBs	Every meaningful word belongs to that one category
Stainless steel sink	Not Steel. Sanitaryware, from the AI's hint	"Stainless" is unknown to the rule, so the rule places nothing; the AI read the whole name
Sand paper	Not Masonry	"Paper" is unknown: the rule places nothing
GI pipe 1 inch	Only by the AI's hint, or none	"Pipe" is in Plumbing, Fire and Gas: a shared word places nothing

The order matters: the AI's hint first (it reads the whole name in context), then the strict rule, then nothing. The AI may only name one of the company's own switched-on categories; it can never create one. This was tightened on 5 Nov after real mistakes ("Stainless steel sink" went to Steel).

6 · WHERE CATEGORIES SHOW UP IN THE PRODUCT

One column, used everywhere

WHERE	WHAT THE USER SEES	BUILT IN
Materials › Items	Items grouped Main › Sub; a picker per item; a "No category" list; a Categories panel to add, rename, reorder, switch off; "Add the standard builder list".	PRC-01, PRC-03
Materials › Stock	A card per main category (items, running low, spend); items grouped by category; category filter.	PRC-17d
Deliveries and use	Every row names its category; filter by category.	PRC-17b
Daily report (DPR)	Materials grouped by main category, in the company's order ("Other" for the rest).	PRC-17c
Excel import	Main category and Sub-category columns.	PRC-05
WhatsApp: set up	"Set up categories" adds the standard list (admin or PM). "What categories do I have?" lists them.	PRC-02c
WhatsApp: new item	The card line shows "📂 Electrical › DB & MCBs" before Save.	PRC-02, PRC-02b
WhatsApp: move an item	"Put wash basin in Sanitaryware" changes its category.	PRC-02b
WhatsApp: ask by category	"Electrical stock?", "What plumbing came this week?"; long answers grouped by main category.	PRC-17a, 17b

The API (what the dashboard and the WhatsApp assistant both call)

ROUTE	WHAT IT DOES
`GET /materials/categories`	The company's tree in order, items per category, and the count with no category.
`POST /materials/categories`	Add a main category, or a sub with `parent_id`.
`PATCH /materials/categories/{id}`	Rename, reorder (`sort_order`), switch off (`is_active`).
`DELETE /materials/categories/{id}`	Delete only if empty; else 409.
`POST /materials/categories/standard`	Add the standard builder list and place items by fixed rules; safe to repeat.
`GET /materials/categories/suggest?name&hint`	The suggestion of part 5, read-only.

ROUTE	WHAT IT DOES
`POST/PATCH /materials` with `category_id`	Put an item in a category, or take it out (null).
`GET /materials/inventory|stock-summary|inflows|outflows?category_id=`	Stock, period stock, deliveries, use filtered by category (main includes subs); rows carry `category` and `main_category`.

The assistant uses the same routes as tools (`material_categories`, `add_standard_categories`, `stock_levels` with `category_id`, `update_material` with a category in words). One rule in one place: the API decides; the dashboard and WhatsApp only show it.

7 · THE CODE: FILES TO READ, IN ORDER

#	FILE	WHAT YOU LEARN THERE
1	`backend/migrations/versions/0503_material_categories.py`	The table, the unique index, row-level security, the new column.
2	`backend/src/mesiri/domains/materials/categories.py`	The standard list; placing rules; `suggest_path`, `read_hint`, `suggest`.
3	`backend/src/mesiri/infrastructure/postgres/repositories/material_categories.py`	The SQL: list with counts, find, in-use check.
4	`backend/src/mesiri/domains/materials/router.py` (search "categories")	The routes and their rules (two levels, 409s, switch off).
5	`backend/src/mesiri/application/materials/categories.py`	Adding the standard list to an existing company.
6	`backend/src/mesiri/domains/materials/stock_summary.py`	Period stock, running low, category filter.
7	`apps/dashboard/src/components/materials/categories-panel.tsx`, `catalogue-view.tsx`, `category-view.tsx`	The dashboard: tree, picker, Stock by category.
8	`apps/whatsapp-assistant/src/agent_api/materials.py` (`settle_category`)	How a WhatsApp card gets its category.
9	`apps/whatsapp-assistant/src/agent_api/tools.py`, `SOP.md` rule 6	What the assistant may call and the rules it follows.
10	`backend/tests/unit/test_material_category_suggest.py`, `backend/tests/contract/test_materials_api_contract.py`	The behaviour, as tests: read these to see every case.

Plans and decisions: `docs/plans/procurement-plan.md` (§4a, §4b, decision D5), tracker rows PRC-01 to PRC-03 and PRC-17a to 17d.

8 · WHAT TO BUILD NEXT

Categories as the backbone for money, buying and control

Today categories sort, filter, total spend and answer questions. The next steps make them drive decisions. Status: Have Partly Missing.

CAPABILITY	PRIORITY	MESIRI	WHAT TO BUILD, AND WHERE
Role check on category and item writes	Must	Partly	Only the dashboard tab is admin-only; the API itself does not check the role (known gap, PRC-01). Add the same check as `may_set_unit_conversions` to the category and catalogue routes. Security-sensitive: needs approval.
Spend by category over time	Should	Partly	Spend per category for a period exists (Stock tab). Add a month-by-month trend per project; feeds the weekly/monthly report (PRC-19).
Move many items at once	Should	Missing	Select several items in Items › No category and place them together; a bulk `PATCH` or a loop in the panel.
Categories on requests, orders and bills	Should	Missing	Comes free with procurement (PRC-06..11): order lines carry the item, so the category; "orders pending by category", "Electrical bills this month".
Suppliers by category	Should	Missing	Which vendors supply which categories, from delivery history; rate comparison within a category (PRC-14, PRC-16).
Category → cost centre	Should	Missing	Map each main category to a cost centre so material cost lands in the right bucket (CC-01) and can be compared with the BOQ section.
Owner per category	Should	Missing	A person responsible for each main category (MEP engineer for Electrical/Plumbing): their alerts, their approvals.
Running-low alerts per category	Should	Partly	Running low exists per item; send it to the category's owner, and allow a minimum level per item.
GST rate and HSN per category	Should	Missing	A default GST rate and HSN code on the category, inherited by its items, overridable per item; needed for bill matching and GST (Accounts).
Wastage allowance per category	Nice	Missing	A default allowance (e.g. Structural 2%) inherited by items; used by reconciliation (CC-08).
Attribute templates per category	Nice	Missing	Steel items have diameter and grade; pipes have size and class; tiles have size and finish. Structured fields instead of putting everything in the name.
Merge two categories	Nice	Missing	Move everything from one into another, then switch the first off.

Suggested order: the role check first (small, closes a real gap), then cost-centre mapping and owner per category (they make categories drive cost control and alerts), then GST/HSN with procurement and bills. The rest follow procurement naturally.

9 · TERMS

Main category The top level: a trade or big group (Electrical).

Sub-category	A group under a main category (Switches & sockets).
Item	One thing bought, counted and used; each size or type its own item.
Standard list	The 7 main and 27 sub-categories every new company starts with.
Switched off	A category no longer offered; items keep it.
No category	Items not yet placed; shown so someone places them.
Hint	The AI's category in words, checked against the company's own list.
HSN code	The tax classification code used on GST bills.
Cost centre	The bucket a cost is reported in (materials, labour, fuel); not the same as a material category.

CHECK YOURSELF

1. Is "TMT bar 12 mm" a category, a sub-category or an item? Why?
2. Why does Mesiri switch a used category off instead of deleting it?
3. Why does "Stainless steel sink" not go to Structural › Steel?
4. A GM asks "how much did we spend on electrical this month?" Which page and which route answer it today?
5. Which next step would let material cost be compared with the electrical part of the BOQ?

Answers: 1. An item (Structural › Steel); sizes are items, not categories. 2. Its items keep their history and totals; it is simply no longer offered. 3. The strict rule places nothing when a word ("stainless") is unknown; the AI's hint, which reads the whole name, says Sanitaryware. 4. Materials › Stock (category cards, this month), from `GET /materials/stock-summary`. 5. Mapping each category to a cost centre, then to its BOQ section (CC-01, CC-08).